

Heterogeneous Learning Mechanisms in Zero-Sum Games: From Best-Iterate to Last-Iterate Convergence

Xinxiang Guo^{1,2†}, Junyue Zhang^{1,2†}, Yifen Mu^{2*†}, Xiao Wang^{3*†},
Ioannis Panageas^{4†}

¹School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing, 100190, China.

²State Key Laboratory of Mathematical Sciences, Chinese Academy of Sciences, Beijing, 100190, China.

³MoE Key Laboratory of Interdisciplinary Research of Computation and Economics, Shanghai University of Finance and Economics, Shanghai, 200433, China.

⁴University of California, Irvine, 92697, USA.

*Corresponding author(s). E-mail(s): mu@amss.ac.cn;
wangxiao@sufe.edu.cn;

Contributing authors: guoxinxiang@amss.ac.cn;
zhangjunyue@amss.ac.cn; ipanagea@ics.uci.edu;

[†]These authors contributed equally to this work.

Abstract

Last-iterate convergence in zero-sum games has received extensive study with many seminal works [1–7]. Nevertheless, all the aforementioned works focus on simultaneous updates, where both players use the same dynamics. Unfortunately, the picture becomes elusive if the players update in an alternating way, using different dynamics, and little is known about provable convergence guarantees. In this paper, we investigate last-iterate convergence guarantees in repeated two-player zero-sum games under heterogeneous updating rules—that is, when the two players use different dynamics. More specifically, we focus on the setting where the row player employs the mirror descent (MD) algorithm, while the column player knows the strategy of the row player and responds with a best response (BR). We show that when the column player plays BR, the row player

achieves best-iterate convergence to their minimax strategy and, under mild conditions, also enjoys last-iterate convergence. Second, when the column player adopts smoothed best response, the stage strategy of the row player exhibits last-iterate convergence. Finally, by using Tikhonov regularization, we prove that the joint strategy profile achieves last-iterate convergence to Nash equilibrium (NE), matching the rate of optimistic and extra-gradient from [5]. We also complement our theoretical results via illustrative simulations. These experiments provide intuitive evidence that heterogeneous learning significantly changes the dynamics in repeated games and enhances the convergence to NE.

Keywords: Game dynamics, Heterogeneous learning, Zero-sum games, Last-iterate convergence

1 Introduction

Learning agents, ranging from AI programs to human decision-makers, continually integrate information from past experience and adapt their behavior in pursuit of improved future performance. In many real-world scenarios, including markets [8, 9], auctions [10, 11], autonomous driving [12], and human society, such agents rarely operate in isolation; instead, they interact strategically with other adaptive agents. These multi-agent interactions are naturally modeled as games, providing a foundational framework for analyzing their strategic behaviors and long-term outcomes. Developing rigorous theoretical guarantees for how learning agents evolve within these game-theoretic environments is essential for understanding, and ultimately designing, complex systems that are reliable, robust, and safe.

A large body of theoretical results establishes convergence guarantees primarily in terms of time-averaged convergence [13, 14], which is closely related to the notion of regret. In particular, it is well known that in zero-sum games, the time average of strategy profiles generated by no-external-regret learners converges to the Nash equilibrium (NE) set, while in general-sum games, such averages converge to the set of coarse correlated equilibria [15].

However, time-averaged convergence does not characterize the day-to-day behavior of agents. In many applications, especially those involving real-time decision-making, the last iterate of learning dynamics is of primary interest. Meanwhile, further studies have revealed that stage strategies generated by classical learning algorithms do not necessarily converge in the last-iterate sense, even in simple two-player zero-sum games. For instance, Mertikoupoulos et al. [16] showed that continuous-time Follow-the-Regularized-Leader (FTRL) dynamics exhibit Poincaré recurrence and [17] demonstrated that when both players employ the multiplicative weights update (MWU) algorithm, the resulting discrete-time dynamics may diverge from the NE. These phenomena help explain the instability commonly observed in the training of generative adversarial networks (GANs) and underscore the necessity of learning algorithms with last-iterate convergence guarantees.

Motivated by this challenge, a growing line of research has proposed optimistic or predictive variants of classical learning algorithms, which incorporate an predicting

term into the update rule. Such algorithms, including optimistic mirror descent (OMD) [18, 19] and optimistic gradient methods [2, 4, 20], have been shown to achieve last-iterate convergence in zero-sum games under self-play.

Notably, all the aforementioned results are derived under the assumption of homogeneous learning, where all agents employ the same learning algorithm. However, learning agents in the real world are often inherently heterogeneous, differing in their decision rules, computational capabilities, and information accessibility [21]. For example, in competitive markets, firms rarely adopt identical pricing algorithms on their respective platforms [8]. Moreover, heterogeneous learning has attracted increasing attention owing to their empirical effectiveness in practical applications like asymmetric training [1]. These observations naturally raise the question:

How does heterogeneity in learning dynamics affect the convergence and stability?

Therefore, it is both natural and necessary to investigate learning dynamics involving heterogeneous agents and to characterize their convergence properties.

In this paper, we investigate the long-run behavior of learning dynamics under a particular yet fundamental class of heterogeneous learning mechanisms in zero-sum games, where one player employs a no-regret learning algorithm, mirror descent (MD), while the opponent plays a best response (BR) to the learner’s current strategy. This setting is of special interest for two reasons. First, taking best responses represents the most adversarial and strategically natural reaction to a no-regret learner, thereby capturing a worst-case interaction scenario. Second, this asymmetric learning structure closely characterizes adversarial training [22–24], and commonly studied two-time-scale learning dynamics [25] can be viewed as its smooth variants. By focusing on this setting, we aim to provide a detailed understanding of the resulting system dynamics and their long-run behavior.

We investigate three distinct heterogeneous learning mechanisms in zero-sum games, each incorporating popular algorithms, best-response strategies, or regularization techniques, and establish their last-iterate convergence. To the best of our knowledge, research on heterogeneous learning in games remains relatively scarce, and its convergence analysis presents notable theoretical challenges. Specifically, we demonstrate the following results.

1. Firstly, we consider the setting in which the row player employs MD and the column player adopts a BR. We prove that the strategy sequence of the row player exhibits best-iterate convergence to their NE strategy. Further analysis shows that when MD instantiates as projected gradient descent (PGD) algorithm, the last-iterate strategy of the row player can be bounded with an error of order $O(T^{-1/2})$.
2. Secondly, we examine the case where the column player adopts the smoothed best response (SBR), while the row player continues to use MD. We establish that the row player achieves last-iterate convergence to an approximate NE strategy, with a linear convergence rate.
3. Thirdly, by incorporating Tikhonov regularization into MD, we prove that the joint strategy profile attains last-iterate convergence to a Nash equilibrium. When MD instantiates as PGD, the convergence rate is $\tilde{O}(T^{-1/2})$, matching the best-known

rate for optimistic algorithms reported in [5]. Our theoretical results are further validated by simulations, which demonstrate that the studied heterogeneous learning mechanisms enhance convergence speed beyond that of optimistic algorithms.

1.1 Related work

Heterogeneous dynamics. Compared with homogeneous learning, the analysis of heterogeneous learning in games remains relatively limited and presents substantially greater challenges. Johanson et al. [26] proposed the CFR-BR algorithm for imperfect-information extensive-form zero-sum games and proved its time-averaged convergence. Arslantas et al. [27] established the convergence of fictitious play with heterogeneous parameters in zero-sum stochastic games. However, these works focus exclusively on time-averaged convergence and do not analyze the resulting day-to-day dynamics. More recently, Bielawski et al. [28] considered settings in which learning agents employ different learning rates and characterized chaotic behavior arising in population games. Collectively, these studies reflect a growing interest in understanding heterogeneous learning dynamics in games.

From a different perspective on heterogeneity, Mu and Guo [29] investigated how to maximally exploit an opponent using a finite-memory strategy and analyzed the dynamic behavior of the resulting optimal strategy. Guo and Mu [30] studied the optimal strategy against the multiplicative weights update (MWU) algorithm and proved the periodicity of the induced dynamics in 2×2 zero-sum games. The opponent in these works pursues a long-term reward, while in our setting the opponent only seeks the instantaneous reward. Most closely related to our work, Guo et al. [31] examined the dynamics arising from the interaction between Hedge and best response in zero-sum games. In contrast, we generalize their results to a broader class of learning algorithms.

Last-iterate convergence. Last-iterate convergence in zero-sum games has received extensive study with many seminal works [2, 7]. [1] formally proved that the last iterate of OMD dynamics converges to an equilibrium in the case of bilinear zero-sum games. [32] proposed a non-asymptotic local convergence theory and proved that OMD exploits the interaction term to achieve exponential convergence to unstable Nash equilibria. [20] proved the last-iterate convergence of extra-gradient, with a rate of $O(T^{-1/2})$, in smooth convex-concave saddle point problems. [2] showed that Optimistic Gradient Descent Ascent (OGDA) exhibits last-iterate convergence to saddle points in unconstrained convex-concave min-max optimization problems. All these convergence results are confined to homogeneous learning settings, whereas we shift our focus to establishing last-iterate convergence under heterogeneous learning dynamics.

Notations Let $\mathbb{R}^n$ be the n -dimensional Euclidean space. Let $\|\cdot\|$ denote the L_2 -norm and $\langle \cdot, \cdot \rangle$ denote the Euclidean inner product. Let $\mathbf{1} \in \mathbb{R}^n$ denote the all-ones vector and $\mathbf{0} \in \mathbb{R}^n$ denote the all-zeros vector. Given a matrix $A \in \mathbb{R}^{n \times m}$, let $\|A\|$ denote the spectral norm of A , i.e., the maximal singular value of A , and let $\|A\|_{\max}$ be the max norm of A , i.e., $\|A\|_{\max} = \max_{i,j} |a_{ij}|$. When Ω is a compact and convex subset of $\mathbb{R}^n$, Proj_{Ω} is the projection operator on Ω , i.e., $\text{Proj}_{\Omega}(x) = \arg \min_{x' \in \Omega} \|x - x'\|^2$. Given a convex function $f : \Omega \rightarrow \mathbb{R}$, $\partial f(x)$ represents the set of subgradients of f at

x , defined to be $\partial f(x) := \{y \in \mathbb{R}^n \mid f(x') \geq f(x) + \langle y, x' - x \rangle, \forall x' \in \Omega\}$. Denote the set $\{0, 1, 2, \dots, T\}$ as $[T]$ and δ_{ij} the Kronecker delta function.

2 Preliminaries

2.1 Two-player Zero-sum Games

Consider a two-player zero-sum game with action sets $\mathcal{I} = \{1, \dots, n\}$ for Player X (the row player) and $\mathcal{J} = \{1, \dots, m\}$ for Player Y (the column player). The payoff matrix $A \in \mathbb{R}^{n \times m}$ specifies the gain a_{ij} for Player Y and the corresponding loss for Player X when actions (i, j) are played.

A mixed strategy for Player X is a probability vector $x \in \Delta(\mathcal{I})$, where $\Delta(\mathcal{I}) = \{x \in \mathbb{R}^n \mid x \geq \mathbf{0}, \mathbf{1}^\top x = 1\}$; similarly, $y \in \Delta(\mathcal{J})$ denotes a mixed strategy for Player Y. Under a strategy profile (x, y) , the payoff for Player Y is $x^\top A y$. In an oracle repeated game, the strategy profile at stage t is denoted by (x_t, y_t) and can be observed. Let $x_{i,t}$ and $y_{j,t}$ represent the i -th and j -th components of x_t and y_t , respectively.

A strategy profile (x, y) is an ε -approximate NE (ε -NE) if

$$\begin{aligned} x^\top A y &\leq (x')^\top A y + \varepsilon, \quad \forall x' \in \Delta(\mathcal{I}), \\ x^\top A y &\geq x^\top A y' - \varepsilon, \quad \forall y' \in \Delta(\mathcal{J}). \end{aligned} \tag{1}$$

An NE, denoted by (x^*, y^*) , is an ε -NE with $\varepsilon = 0$. The value of the zero-sum game, denoted by v^* , is $(x^*)^\top A y^*$. By the minimax theorem [33],

$$v^* = \min_{x \in \Delta(\mathcal{I})} \max_{y \in \Delta(\mathcal{J})} x^\top A y = \max_{y \in \Delta(\mathcal{J})} \min_{x \in \Delta(\mathcal{I})} x^\top A y. \tag{2}$$

We define the value function $V : \Delta(\mathcal{I}) \rightarrow \mathbb{R}$ to be:

$$V(x) = \max_{y \in \Delta(\mathcal{J})} x^\top A y. \tag{3}$$

Hence, v^* is the minimum of V and represents the minimum loss Player X can guarantee in the worst-case scenario. Given a strategy $x \in \Delta(\mathcal{I})$, the quantity $V(x) - v^*$ measures the goodness of x . Due to its resemblance to duality gap, we refer to this quantity as *semi-duality-gap*. Formally, for any $x \in \Delta(\mathcal{I})$, the semi-duality-gap of x is defined to be

$$\text{SDP}(x) = V(x) - v^* = \max_{y \in \Delta(\mathcal{J})} x^\top A y - v^*.$$

2.2 Mirror Descent

Mirror descent [34] is a powerful first-order optimization method that generalizes classical gradient descent to non-Euclidean geometries. It has become a fundamental algorithm in convex optimization, online learning, and game theory [35]. Its update rule is intrinsically linked to the notion of Bregman divergence. Selecting specific divergences recovers two widely studied algorithms: MWU and PGD, as shown below.

Let $\mathcal{X} \subseteq \mathbb{R}^n$ be a convex set. Given any ρ -strongly convex function $\psi : \mathcal{X} \rightarrow \mathbb{R}$, the associated *Bregman divergence* is defined to be

$$D_\psi(x', x) = \psi(x') - \psi(x) - \langle \nabla \psi(x), x' - x \rangle, \quad \forall x, x' \in \mathcal{X}, \quad (4)$$

where we assume that the subdifferential $\partial\psi$ admits a continuous selection $\nabla\psi : \text{dom } \partial\psi \rightarrow \mathbb{R}^n$ with $\nabla\psi(x) \in \partial\psi(x)$ for all $x \in \text{dom } \partial\psi$.

Let $f : \mathcal{X} \rightarrow \mathbb{R}$ be a convex function. The update rule of MD induced by D_ψ to solve a minima of f is given by $x_0 = \arg \min_{x \in \mathcal{X}} \psi(x)$ and

$$x_{t+1} = \arg \min_{x \in \mathcal{X}} \{ \langle f'(x_t), x - x_t \rangle + \lambda_t D_\psi(x, x_t) \}, \quad \forall t \geq 0, \quad (\text{MD})$$

where $f'(x_t) \in \partial f(x_t)$ and $1/\lambda_t$ is called the *learning rate* or *step size*. In this work, we restrict our attention to the case of constant step size, i.e., $\lambda_t = \lambda$, as pointed out by Lin et al. [36], decreasing step size is unnatural from an economic point of view, because it treats newly acquired information with decreasing importance.

For the function ψ , two common choices are as follows.

Example 1 (Euclidean squared norm) When $\mathcal{X}$ is a general convex set, the Euclidean squared norm $\psi(x) = \frac{1}{2}\|x\|^2$ is typically used. This 1-strongly convex function induces the squared Euclidean distance as its Bregman divergence $D(x, y) = \frac{1}{2}\|x - y\|^2$ and MD instantiates as PGD. The update rule (MD) turns to $x_0 = \arg \min_{x \in \mathcal{X}} \|x\|^2$ and

$$x_{t+1} = \text{Proj}_{\mathcal{X}}(x_t - \frac{1}{\lambda_t} f'(x_t)).$$

Example 2 (Negative entropy function) When $\mathcal{X}$ is the n -dimensional simplex Δ_n , the negative entropy function $\psi(x) = \sum_{i=1}^n x_i \log x_i$ naturally fits its geometric structure and generates the Kullback-Leibler (KL) divergence $D_\psi(x, y) = \sum_{i=1}^n x_i \log(x_i/y_i)$ and MD instantiates as the MWU (also called Hedge) algorithm. The update rule (MD) turns to $x_0 = (\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n})$ and

$$x_{i,t+1} = \frac{x_{i,t} e^{-\lambda_t^{-1} [f'(x_t)]_i}}{\sum_{j=1}^n x_{j,t} e^{-\lambda_t^{-1} [f'(x_t)]_j}},$$

where $[f'(x_t)]_i$ represents the i -th element of the vector $f'(x_t)$.

The convergence properties of MD have been extensively studied in the context of convex optimization, under various assumptions on the function f . For more details, we refer readers to see the lecture note [37]. As this note considers a broader scenario, we present the needed results below and defer their proofs in the appendix.

Lemma 1 *Let the sequence $\{x_t\}$ be generated by (MD) with constant $\lambda_t = \lambda$:*

1. If f is L_0 -Lipschitz continuous, i.e., $|f(x) - f(y)| \leq L_0 \|x - y\|$ for all $x, y \in \mathcal{X}$, then

$$\frac{1}{T+1} \sum_{t=0}^T f(x_t) - f(x) \leq \frac{L_0^2}{2\rho\lambda} + \frac{\lambda D_\psi(x, x_0)}{T+1}, \quad \forall x \in \mathcal{X}.$$

2. If f is differentiable with L_1 -Lipschitz continuous gradient, i.e., $\|\nabla f(x) - \nabla f(y)\| \leq L_1 \|x - y\|$ for all $x, y \in \mathcal{X}$, then for any $\lambda \geq L_1/\rho$, we have

$$f(x_t) \leq f(x) + \frac{\lambda D_\psi(x, x_0)}{t}, \quad \forall x \in \mathcal{X}.$$

3. If f is μ -strongly convex and L_1 -smooth with a unique minimizer x^* , then for any $\lambda \geq L_1/\rho$,

$$\|x_t - x^*\|^2 \leq \frac{2\lambda D_\psi(x^*, x_0)}{\mu t}.$$

4. Under the same assumptions in 3, if $\psi(x) = \frac{1}{2}\|x\|^2$, then for any $\lambda \geq (\mu + L_1)/2$,

$$\|x_t - x^*\| \leq \left(1 - \frac{\mu}{\lambda}\right)^t \|x_0 - x^*\|.$$

2.3 Best Response Dynamics

For Player Y (and similarly for Player X), define the set-valued best response mapping BR as

$$\text{BR}(x) = \{y \in \Delta(\mathcal{J}) \mid x^\top A y \geq x^\top A y', \forall y' \in \Delta(\mathcal{J})\}, \quad (5)$$

which is non-empty for all $x \in \Delta(\mathcal{I})$. Taking a best response to the opponent's strategy is a natural response. The continuous-time *best response dynamics* [38] is given by

$$\dot{x} \in \text{BR}(y) - x, \quad \dot{y} \in \text{BR}(x) - y,$$

where $\dot{x}$ and $\dot{y}$ denote the time derivatives of $x(t)$ and $y(t)$, respectively.

And in a discrete-time alternating zero-sum game [39], both players adopt best response dynamics. Given any initial strategy $x_0 \in \Delta(\mathcal{I})$, the update rules are given by

$$y_t \in \text{BR}(x_{t-1}), \quad x_{t+1} \in \text{BR}(y_t), \quad t \geq 1.$$

3 Main Results

We investigate three general and naturally occurring forms of heterogeneous learning dynamics in two-player zero-sum games. By “general and natural,” we refer to settings in which one player employs a classic algorithm, namely MD or its Tikhonov-regularized variant, that is widely applicable across domains and captures the intrinsic incentives of a learning agent, while the opponent adopts a best response or its smoothed variant. We establish convergence guarantees at different levels: from

best-iterate to last-iterate convergence, and from unilateral to bilateral convergence. Together, these results demonstrate the substantial effectiveness of heterogeneous learning mechanisms and reveal the role of regularization in enhancing convergence behavior.

3.1 Unilateral convergence of MD against BR

In this subsection, let Player X employ MD to update their stage strategy and let Player Y respond with a BR to Player X's strategy. The specific update rules are given by: $x_0 = \arg \min_{x \in \Delta(\mathcal{I})} \psi(x)$ and for $t \geq 0$,

$$\begin{aligned} y_t &\in \text{BR}(x_t), \\ x_{t+1} &= \arg \min_{x \in \Delta(\mathcal{I})} \{ \langle Ay_t, x - x_t \rangle + \lambda D_\psi(x, x_t) \}. \end{aligned} \quad (6)$$

A key property of MD is that the Bregman divergence between a point and the iterates before and after an MD update admits a bound. This bound follows from the three-point identity of the Bregman divergence (Lemma 15), which serves as an analogue of the triangle inequality in this context. We present this property in a form tailored to our analysis.

Lemma 2 *Let x^* be an NE strategy of Player X. Then, we have*

$$D_\psi(x^*, x_{t+1}) \leq D_\psi(x^*, x_t) + \frac{1}{\lambda} \langle Ay_t, x^* - x_t \rangle + \frac{\|Ay_t\|^2}{2\rho\lambda^2}. \quad (7)$$

The second term on the right-hand side of (7) is always non-positive, since $x_t^\top Ay_t \geq (x^*)^\top Ay^* \geq (x^*)^\top Ay_t$. Consequently, in the region where this term is sufficiently negative, the Bregman divergence between Player X's strategy and the NE strategy decreases by at least a fixed amount at each stage. It follows that Player X's strategy cannot remain in such a region indefinitely, as formalized in the following theorem.

Theorem 3 *Given a finite repeated time horizon T , let $\{x_t\}_{t=0}^T$ be the strategy sequence generated by (6). Then, the best iterate of $\{x_t\}_{t=0}^T$ satisfies:*

$$\min_{t \in [T]} x_t^\top Ay_t - v^* \leq \frac{n\|A\|_{\max}^2}{2\lambda\rho} + \frac{\lambda D_\psi(x^*, x_0)}{T}, \quad (8)$$

where x^* is an NE strategy of Player X. Furthermore, under the uniform boundedness condition $D_\psi(x, x_0) \leq M_D$ for all $x \in \Delta(\mathcal{I})$, setting $\lambda = \sqrt{\frac{nT\|A\|_{\max}^2}{2\rho M_D}}$ yields the optimal convergence rate:

$$\min_{t \in [T]} x_t^\top Ay_t - v^* \leq \sqrt{\frac{2nM_D\|A\|_{\max}^2}{\rho T}}. \quad (9)$$

Proof of Theorem 3 Denote the region Z_d to be:

$$Z_d = \{x \in \Delta(\mathcal{I}) \mid \max_{y \in \Delta(\mathcal{J})} x^\top Ay - v^* > \frac{n\|A\|_{\max}^2}{2\lambda\rho} + \frac{\lambda D_\psi(x^*, x_0)}{T}\}.$$

If Player X's stage strategies always stay in the region Z_d , then by Lemma 2, it holds for all $1 \leq t \leq T$ that

$$\begin{aligned} D_\psi(x^*, x_t) &\leq D_\psi(x^*, x_{t-1}) + \frac{1}{\lambda} \langle Ay_{t-1}, x^* - x_{t-1} \rangle + \frac{\|Ay_{t-1}\|^2}{2\rho\lambda^2} \\ &\leq D_\psi(x^*, x_{t-1}) + \frac{1}{\lambda} (v^* - x_{t-1}^\top Ay_{t-1}) + \frac{n\|A\|_{\max}^2}{2\rho\lambda^2} \\ &< D_\psi(x^*, x_{t-1}) - \frac{1}{\lambda} \left(\frac{n\|A\|_{\max}^2}{2\lambda\rho} + \frac{\lambda D_\psi(x^*, x_0)}{T} \right) + \frac{n\|A\|_{\max}^2}{2\rho\lambda^2} \\ &< D_\psi(x^*, x_{t-1}) - \frac{D_\psi(x^*, x_0)}{T} \end{aligned}$$

Summing this inequality over from $t = 1$ to T leads to $D_\psi(x^*, x_T) < 0$, which contradicts to $D_\psi(x^*, x_T) \geq 0$. This completes the proof. The latter result follows immediately by applying the basic inequality to (8). $\square$

This result demonstrates that by the end of the repeated game, there exists a stage at which Player X's strategy is approximately as effective as an NE strategy. This guarantee is known as best-iterate convergence. Similar guarantees have been established in other works, such as Lockhart et al. [40] and Lin et al. [41]. However, those studies consider decreasing learning rates, which in our setting would enable last-iterate convergence instead.

Theorem 3 can also be proven using Lemma 1, following an approach similar to the analysis of the subsequent theorems. It is straightforward to verify that $V(x)$ is convex and Lipschitz continuous (Lemma 16 in the Appendix). For any $y \in \text{BR}(x)$, the vector Ay is a subgradient of V at x (Lemma 17 in the Appendix). Substituting these properties directly into Lemma 1 completes the proof. For different function ψ , we have the following corollaries.

Corollary 4 *For the Hedge algorithm, i.e., $\psi(x) = \sum_{i=1}^n x_i \ln x_i$, we have $\rho = 1$ and $D_\psi(x, x_0) \leq \ln n$ for all $x \in \Delta(\mathcal{I})$. Then, by taking $\lambda = \sqrt{\frac{nT\|A\|_{\max}^2}{2 \ln n}}$,*

$$\min_{t \in [T]} x_t^\top Ay_t - v^* \leq \sqrt{\frac{2n \ln n \|A\|_{\max}^2}{T}}.$$

Corollary 5 *For the projected gradient descent algorithm, i.e., $\psi(x) = \frac{1}{2}\|x\|^2$, we have $\rho = 1$ and $D_\psi(x, x_0) \leq \frac{1}{2}$ for all $x \in \Delta(\mathcal{I})$. Then, by taking $\lambda = \sqrt{nT\|A\|_{\max}^2}$,*

$$\min_{t \in [T]} x_t^\top Ay_t - v^* \leq \sqrt{\frac{n\|A\|_{\max}^2}{T}}.$$

Best-iterate convergence should not be interpreted as a satisfactory characterization of day-to-day dynamics, as it provides no guarantee regarding the behavior after

the best stage is reached or the property of the final strategy. Intuitively, however, inequality (7) suggests that Player X's strategy cannot drift arbitrarily far from the NE. Guo et al. [42] proved that the Bregman divergence between Player X's strategies and the NE remains bounded when using the Hedge algorithm, under the assumption of a unique interior NE. In this work, we generalize that result to projected gradient descent algorithm.

Before presenting this result, we state a lemma that bounds the distance between the NE strategy and the strategy with a bounded semi-duality-gap.

Lemma 6 *Suppose that the NE of the game is unique. Then, there exists a constant $c > 0$ determined by the payoff matrix A such that for any $x \in \Delta(\mathcal{I})$,*

$$\|x - x^*\| \leq c(V(x) - v^*),$$

where x^* is the unique NE strategy of Player X.

Proof of Lemma 6 By the minimax theorem (2), the NE strategy x^* satisfies that:

$$A^\top x^* \leq v^* \mathbf{1}, \mathbf{1}^\top x^* = 1, x^* \geq \mathbf{0}.$$

Directly using the result (Lemma 18 in the Appendix) from Hoffman [43] yields that there exists a constant $c > 0$ such that for all $x \in \Delta(\mathcal{I})$,

$$\|x - x^*\| \leq c \|A^\top x - v^* \mathbf{1}\|_\infty = c(V(x) - v^*).$$

□

Theorem 7 *Suppose that the NE of the game is unique and Player X employs the projected gradient descent algorithm. Setting $\lambda = \sqrt{nT} \|A\|_{\max}^2$ yields:*

$$V(x_T) - v^* \leq \sqrt{(nc^2 \|A\|_{\max}^2 + 1) \frac{n \|A\|_{\max}^2}{T}}, \quad (10)$$

where $c > 0$ is a constant determined by A .

Proof of Theorem 7 By Theorem 3, denote the best stage by t_0 , i.e., $x_{t_0}^\top A y_{t_0} - v^* \leq \sqrt{\frac{n \|A\|_{\max}^2}{T}}$. For simplicity, we write $\sqrt{\frac{n \|A\|_{\max}^2}{T}}$ as ε_T .

Define the region Z_u to be:

$$Z_u \triangleq \{x \in \Delta(\mathcal{I}) \mid V(x) \leq v^* + \varepsilon_T\} = \{x \in \mathbb{R}^n \mid A^\top x \leq (v^* + \varepsilon_T) \mathbf{1}, \mathbf{1}^\top x = 1, x \geq \mathbf{0}\}. \quad (11)$$

Thus, $x_{t_0} \in Z_u$. By Lemma 6, we obtain that for all $x \in Z_u$:

$$\|x - x^*\| \leq c \varepsilon_T.$$

Now, consider the stage strategies x_t of Player X after t_0 . If $x_t \in Z_u$, then $\|x_t - x^*\| \leq c \varepsilon_T$; otherwise, either $x_{t-1} \in Z_u$ or $x_{t-1} \notin Z_u$. For the former case, by Lemma 2, $D_\psi(x^*, x_t) \leq D_\psi(x^*, x_{t-1}) + \frac{1}{2T} \leq \frac{c^2 \varepsilon_T^2}{2} + \frac{1}{2T}$. For the later case, $D_\psi(x^*, x_t) \leq D_\psi(x^*, x_{t-1}) \leq D_\psi(x^*, x_{t-k})$, where $x_{t-k} \in Z_u$ for some $k > 1$.

All together, we have:

$$D_\psi(x^*, x_t) \leq \frac{c^2 \varepsilon_T^2}{2} + \frac{1}{2T}, \quad \forall t \geq t_0.$$

Since $V(x)$ is Lipschitz continuous, we finally have:

$$\begin{aligned} V(x_T) - v^* &\leq \sqrt{n} \|A\|_{\max} \|x_T - x^*\| = \|A\|_{\max} \sqrt{2n D_\psi(x^*, x_T)} \\ &\leq \sqrt{(nc^2 \|A\|_{\max}^2 + 1) \frac{n \|A\|_{\max}^2}{T}}. \end{aligned}$$

□

Remark 1 The assumption of a unique NE is generic, as the set of zero-sum games with non-unique equilibria has Lebesgue measure zero [17, 44]. In contrast to the requirement in [31], this unique NE need not be interior. Moreover, our result can be extended to other MD algorithms that satisfy the following quadratic growth condition: there exists a constant $L_D > 0$ such that $D_\psi(x, x') \leq L_D \|x - x'\|^2$ for all x, x' in the domain.

In contrast to the non-convergence observed under homogeneous MD [16], our heterogeneous learning mechanism ensures convergence. The key reason lies in the robustness of no-regret learning: such algorithms are designed to guarantee no regret against any adversary, and a best-responding player represents a particularly strong adversarial instance.

The alternating interaction structure inherently places Player X at a disadvantage. Despite this asymmetry, Theorem 7 establishes that by employing MD, Player X can still learn a strategy arbitrarily close to an NE strategy. This result, to a considerable extent, provides a theoretical safety guarantee for applying MD in adversarial decision-making scenarios.

3.2 Unilateral last-iterate convergence of MD against SBR

In this subsection, we retain the previous setting but let Player Y adopt the smoothed best response (SBR) [45] (also known as quantal response in [46]) to the stage strategy of Player X. The update rules are given by: $x_0 = \arg \min_{x \in \Delta(\mathcal{I})} \psi(x)$ and, for $t \geq 0$,

$$\begin{aligned} y_t &= \text{SBR}(x_t; h, \tau) \triangleq \arg \max_{y \in \Delta(\mathcal{J})} \{ \langle Ay, x \rangle - \tau h(y) \}, \\ x_{t+1} &= \arg \min_{x \in \Delta(\mathcal{I})} \{ \langle Ay_t, x - x_t \rangle + \lambda D_\psi(x, x_t) \}, \end{aligned} \tag{12}$$

where $\tau > 0$ is called the *smoothing parameter* and $h(y)$ is a bounded and K -strongly convex function on $\Delta(\mathcal{J})$. Denote the upper bound of $|h|$ on $\Delta(\mathcal{J})$ by M_h . Unlike standard BR, the response function SBR in (12) is single-valued. For simplicity, we denote $\text{SBR}(x; h, \tau)$ by $y(x; \tau)$. Concrete examples can be found in the Appendix.

With a slight abuse of notation, we define the value function $V : \Delta(\mathcal{I}) \rightarrow \mathbb{R}$ to be

$$V(x; \tau) := \max_{y \in \Delta(\mathcal{J})} \{ x^\top Ay - \tau h(y) \}, \tag{13}$$

and $X^*(\tau) = \arg \min_{x \in \Delta(\mathcal{I})} V(x; \tau)$.

First, by the envelope theorem [47], we directly show that $Ay(x; \tau)$ corresponds precisely to the gradient of $V(x; \tau)$ and then prove the convexity of $V(x; \tau)$.

Lemma 8 *The gradient of $V(x; \tau)$ is $Ay(x; \tau)$, and $V(x; \tau)$ is convex.*

Given that $Ay_t = Ay(x_t; \tau)$ is the gradient $\nabla_x V(x_t; \tau)$, Lemma 1 implies that if $V(x; \tau)$ is smooth, the sequence $\{V(x_t; \tau)\}$ converges to the minimum of $V(x; \tau)$. The following lemma will establish the smoothness of $V(x; \tau)$ by showing the Lipschitz continuity of SBR with respect to x , which is different from the BR mapping.

Lemma 9 *The SBR $y(x; \tau)$ with K -strongly convex function h is $\frac{\|A\|}{\tau K}$ -Lipschitz continuous. Further, if $h(y)$ is of the format that $h(y) = \sum_{j=1}^m h_j(y_j)$, where each h_j is smooth and strongly convex and $y(x; \tau)$ lies in the interior of $\Delta(\mathcal{J})$ for all $x \in \Delta(\mathcal{I})$, $y(x; \tau)$ is $\frac{\|A\| M_{\tilde{h}}}{\tau}$ -Lipschitz continuous and differentiable, where*

$$M_{\tilde{h}} = \max_{1 \leq j \leq m} \sup_{x \in \Delta(\mathcal{I})} (h_j''(y_j(x; \tau)))^{-1}.$$

The differential is

$$dy(x; \tau) = \frac{1}{\tau} X A^\top dx,$$

where $X_{ij} = \delta_{ij} / \tilde{h}_i - (\tilde{h}_i \tilde{h}_j \sum_{k=1}^m \tilde{h}_k^{-1})^{-1}$ and $\tilde{h}_j = h_j''(y_j)$.

Corollary 10 *The function $V(x; \tau)$ is $\frac{mn\|A\|_{\max}^2}{\tau K}$ -smooth.*

The next two examples show results for different choices of h .

Example 3 When $h(y) = \frac{1}{2} \|y\|_2^2 = \frac{1}{2} \sum_{j=1}^m y_j^2$, which is 1-strongly convex, the corresponding value function $V(x; \tau)$ is $\frac{mn\|A\|_{\max}^2}{\tau}$ -Lipschitz smooth.

Example 4 (Logistic Quantal Response) When $h(y) = \sum_{j=1}^m y_j \log y_j$, the SBR takes the form:

$$y_{j,t} = \frac{\exp(\tau^{-1} x_t^\top A e_j)}{\sum_{k=1}^m \exp(\tau^{-1} x_t^\top A e_k)}, \quad \forall j = 1, 2, \dots, m, \quad (14)$$

known as the logistic quantal response in [46]. Note that $h_j''(y_j) = y_j^{-1}$ and y_t satisfies that

$$\left(1 + \frac{m-1}{\exp(-\delta/\tau)}\right)^{-1} \leq y_{j,t} \leq \left(1 + \frac{m-1}{\exp(\delta/\tau)}\right)^{-1},$$

where $\delta = \max_{i,j} a_{ij} - \min_{i,j} a_{ij}$. These bounds ensure that y_t remains in the interior of $\Delta(\mathcal{J})$ and is differentiable. Consequently, $V(x; \tau)$ is L_τ -smooth with

$$L_\tau = \frac{mn\|A\|_{\max}^2}{\tau(1 + (m-1)\exp(-\delta/\tau))}.$$

While the convexity and smoothness of $V(x; \tau)$ guarantee the convergence of the sequence $\{V(x_t; \tau)\}$, establishing how the sequence $\{V(x_t)\}$ relates to the original game value v^* requires analyzing the connection between $V(x; \tau)$ and $V(x)$, as shown by the following two lemmas.

Lemma 11 For any $x_\tau^* \in X^*(\tau)$ and $x \in \Delta(\mathcal{I})$, we have

$$\begin{aligned} 0 &\leq V(x_\tau^*) - v^* \leq 2\tau M_h, \\ |V(x; \tau) - V(x)| &\leq \tau M_h. \end{aligned}$$

Consequently, for the sequence $\{x_t\}$ generated by (12), their values $\{V(x_t)\}$ converge to the value v^* of the original game.

Theorem 12 For the sequence $\{x_t\}$ generated by (12) with K -strongly convex h , ρ -strongly convex ψ , and learning rate $\lambda \geq \frac{\|A\|^2}{\tau \rho K}$, the following bound holds:

$$V(x_t) - v^* \leq \frac{\lambda D_\psi(X_\tau^*, x_0)}{t} + 4\tau M_h,$$

where $D_\psi(X_\tau^*, x_0) := \inf_{x_\tau^* \in X_\tau^*} D_\psi(x_\tau^*, x_0)$ is the minimal Bregman divergence between the optimal solution set $X_\tau^* = \arg \min_{x \in \Delta(\mathcal{I})} V(x; \tau)$ and the initial point x_0 .

Proof of Theorem 12 Since $Ay_t = \nabla_x V(x_t; \tau)$ and by Lemma 9, $V(x; \tau)$ is $\frac{\|A\|^2}{\tau K}$ -smooth, applying Lemma 1 with $x = x_\tau^* \in X^*(\tau)$ yields

$$0 \leq V(x_t; \tau) - V(x_\tau^*; \tau) \leq \frac{\lambda D_\psi(x_\tau^*, x_0)}{t}.$$

Furthermore, Lemma 11 imply that

$$|V(x_\tau^*; \tau) - v^*| \leq |V(x_\tau^*; \tau) - V(x_\tau^*)| + |V(x_\tau^*) - v^*| \leq 3\tau M_h.$$

Combining these results, we obtain that

$$V(x_t) - v^* = (V(x_t) - V(x_t; \tau)) + (V(x_t; \tau) - V(x_\tau^*; \tau)) + (V(x_\tau^*; \tau) - v^*) \leq \frac{\lambda D(x_\tau^*, x_0)}{t} + 4\tau M_h.$$

Taking the infimum over all $x_\tau^* \in X^*(\tau)$ yields the result. $\square$

Remark 2 Theorem 12 establishes the linear convergence rate toward an approximate NE strategy, where the approximation error $4\tau M_h$ cannot be eliminated through iteration. However, for a given repeated game horizon T , by choosing $\lambda = \frac{\|A\|^2}{\tau \rho K}$ and setting $\tau = \frac{\|A\|}{2} \sqrt{\frac{M_D}{\rho K M_h T}}$, we obtain the bound

$$V(x_t) - v^* \leq 4\|A\| \sqrt{\frac{M_D M_h}{\rho K T}}.$$

This matches the rate of optimistic and extra-gradient from [5]. Besides, compared to [41] and [40], this result provides last-iterate convergence rather than best-iterate convergence, and applies to a broader class of algorithms, including gradient descent considered in these two papers as a special case.

Note that only unilateral convergence is established for (12). To obtain an NE strategy for Player Y, it is necessary to swap the update roles of Player X and Player Y. This step is required because, in general games, the SBR to an NE strategy does not necessarily constitute an NE strategy itself.

3.3 Bilateral Convergence via Tikhonov Regularization

For the payoff matrix A and the BR mapping, it can be verified that any ϵ -approximate solution (x, y) to the following quasi-variational inequality

$$\begin{aligned} \min_{x' \in \Delta(\mathcal{I})} \langle Ay, x' - x \rangle &\geq 0, \\ \text{s.t. } x &\in \Delta(\mathcal{I}), \quad y \in \text{BR}(x), \end{aligned} \quad (15)$$

forms an ϵ -Nash equilibrium. This characterization motivates the application of Tikhonov regularization [48], a technique originally developed for ill-posed inverse problems and later applied into solving monotone variational inequality problems. The approach is to transform the original problem into a new problem with better properties by adding small regularization.

In this subsection, let Player X adopt Tikhonov-regularized MD and Player Y still takes SBR at each stage. Specifically, the update rules are given by: $x_0 = \arg \min_{x \in \Delta(\mathcal{I})} \psi(x)$ and, for $t \geq 0$,

$$\begin{aligned} y_t &= \arg \max_{y \in \Delta(\mathcal{J})} \{\langle Ay, x_t \rangle - \tau h(y)\}, \\ x_{t+1} &= \arg \min_{x \in \Delta(\mathcal{I})} \{\langle Ay_t + \epsilon x_t, x - x_t \rangle + \lambda D_\psi(x, x_t)\}, \end{aligned} \quad (16)$$

where $\epsilon > 0$ is called the *regularization parameter*. Compared with the update rules (12), Tikhonov regularization incorporates the current x_t as part of the “gradient information” to determine the next x_{t+1} .

Similar to the analysis in Subsection 3.2, we first define the function

$$V(x; \epsilon, \tau) := \max_{y \in \Delta(\mathcal{J})} \left\{ x^\top Ay + \frac{\epsilon}{2} \|x\|^2 - \tau h(y) \right\}.$$

It is straightforward to show that $V(x; \epsilon, \tau)$ is strongly convex and smooth, with its gradient given by $Ay(x; \tau) + \epsilon x$ (See Lemma 20 in the Appendix). Since $\Delta(\mathcal{I})$ is a bounded, closed, and convex set, the minimizer of $V(x; \epsilon, \tau)$ exists and is unique, which we denote by $x^*(\epsilon, \tau)$. Moreover, analogous to $V(x; \epsilon, \tau)$ and $V(x)$, as well as the gap between $V(x^*(\epsilon, \tau))$ and v^* (See Lemma 21 in the Appendix). The following theorem demonstrates the the last-iterate convergence of the joint strategy sequence.

Theorem 13 *For the sequence $\{x_t\}$ generated by (16) with a K -strongly convex function h , a ρ -strongly convex ψ , a constant parameter ϵ , and a constant $\lambda \geq (\frac{\|A\|^2}{\tau K} + \epsilon)/\rho$, we have:*

$$V(x_t) - V(x^*) \leq \frac{\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{t} + 4\tau M_h + \epsilon.$$

More importantly, the strategy profile (x_t, y_t) forms a $C'(t; \epsilon, \tau)$ -approximate NE of the original game, where $C'(t; \epsilon, \tau) = 2 \left(\frac{\|A\|^2}{\tau K} + \epsilon \right) \sqrt{\frac{2\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{\epsilon t}} + 2\tau M_h + \epsilon$.

Proof of Theorem 13 By the convergence results (Lemma 1) of MD, the stage strategy sequence of Player X satisfies:

$$V(x_t; \epsilon, \tau) - V(x^*(\epsilon, \tau); \epsilon, \tau) \leq \frac{\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{t}. \quad (17)$$

Combining this with Lemma 21, we obtain

$$\begin{aligned} & V(x_t) - v^* \\ &= V(x_t) - V(x_t; \epsilon, \tau) + V(x_t; \epsilon, \tau) - V(x^*(\epsilon, \tau); \epsilon, \tau) \\ & \quad + V(x^*(\epsilon, \tau); \epsilon, \tau) - V(x^*(\epsilon, \tau)) + V(x^*(\epsilon, \tau)) - v^* \\ &\leq 4\tau M_h + \epsilon + V(x_t; \epsilon, \tau) - V(x^*(\epsilon, \tau); \epsilon, \tau) \\ &\leq 4\tau M_h + \epsilon + \frac{\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{t}. \end{aligned}$$

For the convergence of the strategy profile (x_t, y_t) , on the one hand, for any $y \in \Delta(\mathcal{J})$, we have

$$\begin{aligned} & x_t^\top A y_t - x_t^\top A y \\ &= x_t^\top A y_t - \tau h(y_t) - x_t^\top A y + \tau h(y) + \tau h(y_t) - \tau h(y) \\ &\geq \tau h(y_t) - \tau h(y) \\ &\geq -2\tau M_h. \end{aligned}$$

On the other hand, by the optimality condition satisfied by $x^*(\epsilon, \tau)$, we have

$$\langle \nabla_x V(x^*(\epsilon, \tau); \epsilon, \tau), x - x^*(\epsilon, \tau) \rangle \geq 0, \quad \forall x \in \Delta(\mathcal{I}).$$

Since $V(x; \epsilon, \tau)$ is ϵ -strongly convex, by Lemma 1: $\|x_t - x^*(\epsilon, \tau)\| \leq \sqrt{\frac{2\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{\epsilon t}}$.

For $\nabla_x V(x_t; \epsilon, \tau)$, we have

$$\begin{aligned} & \langle \nabla_x V(x_t; \epsilon, \tau), x - x_t \rangle \\ &= \langle \nabla_x V(x_t; \epsilon, \tau) - \nabla_x V(x^*(\epsilon, \tau); \epsilon, \tau), x - x_t \rangle + \underbrace{\langle \nabla_x V(x^*(\epsilon, \tau); \epsilon, \tau), x - x^*(\epsilon, \tau) \rangle}_{\geq 0} \\ & \quad + \langle \nabla_x V(x^*(\epsilon, \tau); \epsilon, \tau), x^*(\epsilon, \tau) - x_t \rangle \\ &\geq \underbrace{\langle \nabla_x V(x_t; \epsilon, \tau) - \nabla_x V(x^*(\epsilon, \tau); \epsilon, \tau), x - x_t \rangle}_{\text{term 1}} + \underbrace{\langle \nabla_x V(x^*(\epsilon, \tau); \epsilon, \tau), x^*(\epsilon, \tau) - x_t \rangle}_{\text{term 2}}. \end{aligned}$$

For **term 1**, we have that

$$\begin{aligned} \text{term 1} &= \langle \nabla_x V(x_t; \epsilon, \tau) - \nabla_x V(x^*(\epsilon, \tau); \epsilon, \tau), x - x_t \rangle \\ &\geq -\|\nabla_x V(x_t; \epsilon, \tau) - \nabla_x V(x^*(\epsilon, \tau); \epsilon, \tau)\| \|x - x_t\| \\ &\geq -L_2 \|x_t - x^*(\epsilon, \tau)\| \|x - x_t\| \\ &\geq -2L_2 \|x_t - x^*(\epsilon, \tau)\|, \end{aligned}$$

where the first and the second inequality follow by the Cauchy-Schwarz inequality and the smoothness of $V(x; \epsilon, \tau)$, respectively.

For **term 2**, since $V(x; \epsilon, \tau)$ is ϵ -strongly convex, we obtain that $V(x_t; \epsilon, \tau) \geq V(x^*(\epsilon, \tau); \epsilon, \tau) + \langle \nabla_x V(x^*(\epsilon, \tau); \epsilon, \tau), x_t - x^*(\epsilon, \tau) \rangle + \frac{\epsilon}{2} \|x_t - x^*(\epsilon, \tau)\|^2$. Therefore,

$$\begin{aligned} \text{term 2} &= \langle \nabla_x V(x^*(\epsilon, \tau); \epsilon, \tau), x^*(\epsilon, \tau) - x_t \rangle \\ &\geq V(x^*(\epsilon, \tau); \epsilon, \tau) - V(x_t; \epsilon, \tau) + \frac{\epsilon}{2} \|x_t - x^*(\epsilon, \tau)\|^2 \\ &\geq -\frac{\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{t} + \frac{\epsilon}{2} \|x_t - x^*(\epsilon, \tau)\|^2. \end{aligned}$$

Combining the estimates for **term 1** and **term 2**, we have that

$$\begin{aligned}
& \langle \nabla_x V(x_t; \epsilon, \tau), x - x_t \rangle \\
& \geq \text{term 1} + \text{term 2} \\
& \geq -2L_2 \|x_t - x^*(\epsilon, \tau)\| - \frac{\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{t} + \frac{\epsilon}{2} \|x_t - x^*(\epsilon, \tau)\|^2 \\
& \geq -2L_2 \sqrt{\frac{2\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{\epsilon t}} - \frac{2\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{\epsilon t} + \frac{2\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{\epsilon t} \\
& = -2L_2 \sqrt{\frac{2\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{\epsilon t}},
\end{aligned}$$

where the final inequality holds because for sufficiently large t , we have

$$\|x_t - x^*(\epsilon, \tau)\| \leq \sqrt{\frac{2\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{\epsilon t}} < \frac{2L_2}{\epsilon},$$

which implies that the expression $\frac{\epsilon}{2} \|x_t - x^*(\epsilon, \tau)\|^2 - 2L_2 \|x_t - x^*(\epsilon, \tau)\|$ attains its minimum at $\|x_t - x^*(\epsilon, \tau)\| = \sqrt{\frac{2\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{\epsilon t}}$. Therefore, for any $x \in \Delta(\mathcal{I})$, we have that

$$\begin{aligned}
& x_t^\top A y_t - x^\top A y_t \\
& = -\langle A y_t + \epsilon x_t, x - x_t \rangle + \epsilon(x_t^\top x - x_t^\top x_t) \\
& = -\langle \nabla_x V(x_t; \epsilon, \tau), x - x_t \rangle + \epsilon(x_t^\top x - x_t^\top x_t) \\
& \leq 2L_2 \sqrt{\frac{2\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{\epsilon t}} + \epsilon.
\end{aligned}$$

In conclusion, for the strategy pair (x_t, y_t) , we have

$$\begin{aligned}
x_t^\top A y_t - x^\top A y_t & \leq 2L_2 \sqrt{\frac{2\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{\epsilon t}} + \epsilon, \quad \forall x \in \Delta(\mathcal{I}), \\
x_t^\top A y_t - x_t^\top A y & \geq -2\tau M_h, \quad \forall y \in \Delta(\mathcal{J}).
\end{aligned}$$

Therefore, the profile (x_t, y_t) is a $\left(2L_2 \sqrt{\frac{2\lambda D_\psi(x^*(\epsilon, \tau), x_0)}{\epsilon t}} + 2\tau M_h + \epsilon\right)$ -approximate NE. $\square$

Remark 3 The update rules can be interpreted as Player X employing MD and Player Y taking BR in a newly defined game with two-side regularization. Unlike most of the previous literature, the two players are allowed to use different regularizers. One of the key challenges in applying regularization techniques lies in bounding the induced approximation error, which, in our framework, can be handled easily, even when using the negative entropy function.

Corollary 14 Based on Lemma 1, by taking $\psi(x) = \frac{1}{2} \|x\|^2$ (i.e., MD reduces to the projected gradient descent algorithm) and letting $\lambda \geq \frac{\|A\|^2}{2\tau K} + \epsilon$, we have:

$$V(x_t) - v^* \leq 2\sqrt{n} \|A\|_{\max} \left(1 - \frac{\epsilon}{\lambda}\right)^t + 4\tau M_h + \epsilon.$$

For a given sufficiently large T , by taking h as Euclidean regularizer and setting $\lambda = \frac{\|A\|^2}{2\tau} + \epsilon$, the profile (x_T, y_T) forms a $C'(T)$ -approximate NE, where

$$C'(T) \leq \max\left\{2\left(\frac{2\|A\|^2}{\tau} + \|A\| + \epsilon\right)\left(1 + \frac{2\tau\epsilon}{\|A\|^2}\right)^{-T} + \epsilon, 2\tau\right\}.$$

By choosing $\epsilon = \tau = \sqrt{\log T / T}$, we obtain $C'(T) = \tilde{O}(T^{-1/2})$.

This rate matches that of the optimistic and extra-gradient methods in [5], which is known to be the best convergence rate. Therefore, heterogeneous learning combined with smoothing techniques and Tikhonov regularization provides a competitive approach for computing equilibria.

4 Simulations Results

4.1 Evolutionary trajectory

In this section, we analyze the evolutionary trajectory of Player X's stage strategies under different learning dynamics. These include MD under self-play, alternating MD, MD against BR, and MD against SBR. Here we consider two most popular MD algorithms: Hedge [49] and projected gradient descent (PGD) [50].

We examine the 3×3 matrix game presented in [42], with the payoff matrix:

$$A = \begin{pmatrix} -2 & 1 & 3 \\ 1 & 2 & -2 \\ 2 & 0 & -1 \end{pmatrix}.$$

This game has a unique interior NE: $((\frac{3}{8}, \frac{1}{24}, \frac{7}{12}), (\frac{3}{8}, \frac{1}{3}, \frac{7}{24}))$.

We set the time horizon T to 2000, the learning rate $1/\lambda$ to 0.005, and the parameter τ to 0.1. The function $h(y)$ is chosen as the negative entropy, i.e., $h(y) = -\sum_{j=1}^m y_j \ln y_j$. Figure 1 illustrates the evolutionary trajectories, where the x- and y-axes represent the first and second components of Player X's strategy, respectively.

The first two figures demonstrate that both simultaneous and alternating MD fail to converge to the NE strategy. In contrast, when MD is played against BR, Player X's strategy quickly enters a cycle, with certain iterates lying close to the NE. Replacing BR with SBR allows the strategy to escape the cycle and exhibit strong convergence to the NE. A similar phenomenon is observed for PGD (See Figure 2). These results highlight that heterogeneous learning mechanisms significantly alter the dynamics and promote convergence to NE.

Then, we conduct the same experiments on PGD and analyze the evolutionary trajectory of Player X's strategy sequence. The time horizon is still set 2000, the learning rate is set 0.01, and the parameter τ to 0.1. The function $h(y)$ is also chosen as the negative entropy function. Figure 2 illustrates the evolutionary trajectories, where the x- and y-axes represent the first and second components of Player X's strategy, respectively.

The first two figures show that PGD fails to converge to the NE in both the simultaneous and alternating settings. In contrast, when PGD is played against BR, Player X's strategy quickly falls into a cycle, with some iterates approaching the NE. Replacing BR with SBR enables the strategy to escape this cycle and converge strongly to the NE.

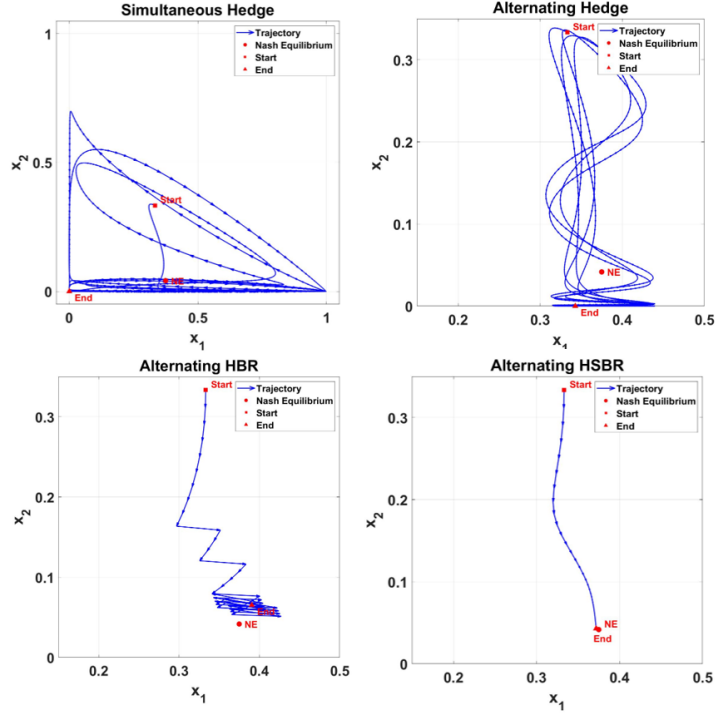


Fig. 1 The evolutionary trajectory of the stage strategy of Player X with Hedge.

4.2 Convergence rate

In this part, we evaluate the convergence performance of several NE-solving paradigms possessing last-iterate convergence. The paradigms under consideration include Optimistic Hedge under self-play (OHSP), Hedge against SBR (HSBR), Optimistic Gradient Descent Ascent (OGDA), and PGD against SBR (PGDSBR). To facilitate a fair comparison, we tune the learning rates and other relevant parameters to optimize convergence speed for each paradigm.

We begin by randomly generating a game matrix $A \in \mathbb{R}^{100 \times 100}$, where each entry is drawn uniformly from the interval $[-1, 1]$, followed by the addition of uniform noise in the range $[-0.1, 0.1]$. This matrix serves as the payoff matrix for evaluating four paradigms for solving Nash equilibria. Each paradigm generates a strategy sequence of Player X. For each strategy x_t , we compute its semi-duality-gap.

The repeated game is simulated over a fixed horizon, and at each iteration, we record the historical semi-duality-gap. This entire procedure is repeated across 50 independent trials, resulting in 50 distance trajectories. We compute the mean of these trajectories at each iteration step. The aggregated results are presented in Figure 3. From the figure, we observe that compared with optimistic variants, especially OHSP, HSBR and PGDSBR demonstrate a faster convergence rate and greater stability.

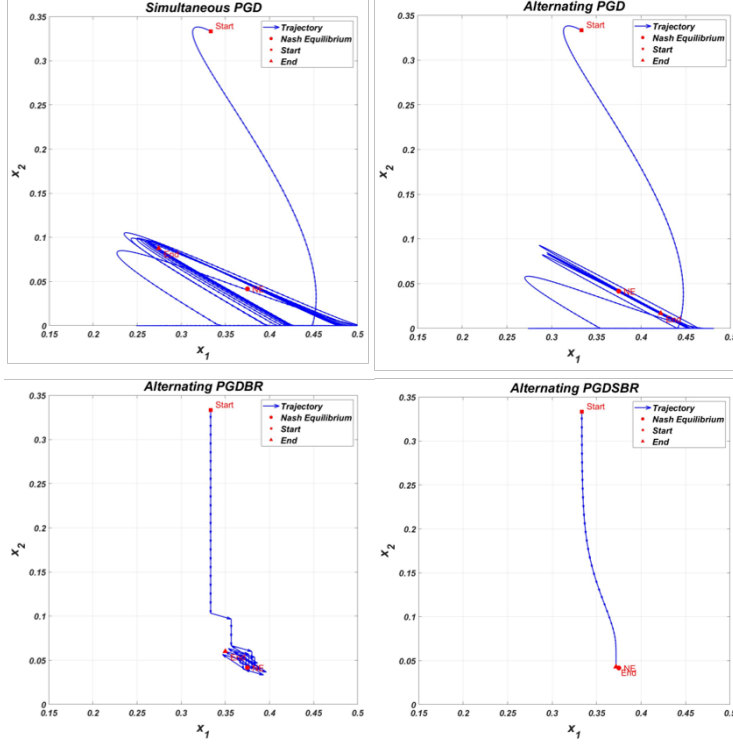


Fig. 2 The evolutionary trajectory of the stage strategy of Player X with PGD.

5 Conclusion and Future Work

In this work, we investigate the alternating interaction between heterogeneous learning dynamics in repeated two-player zero-sum games. We first show that when the row player employs MD and the column player plays a BR, the MD dynamics achieve best-iterate convergence to the minimax strategy. When the column player adopts SBR, we establish last-iterate convergence to the minimax strategy. Further, by incorporating Tikhonov regularization into MD, we prove the bilateral last-iterate convergence to an approximate NE, with a provably linear rate. Our simulations demonstrate that heterogeneous dynamics, particularly MD against SBR, accelerate convergence and improve stability compared to standard optimistic MD. An important direction for future work is to investigate the dynamics in more complex settings, such as multi-player general-sum games and bandit information.

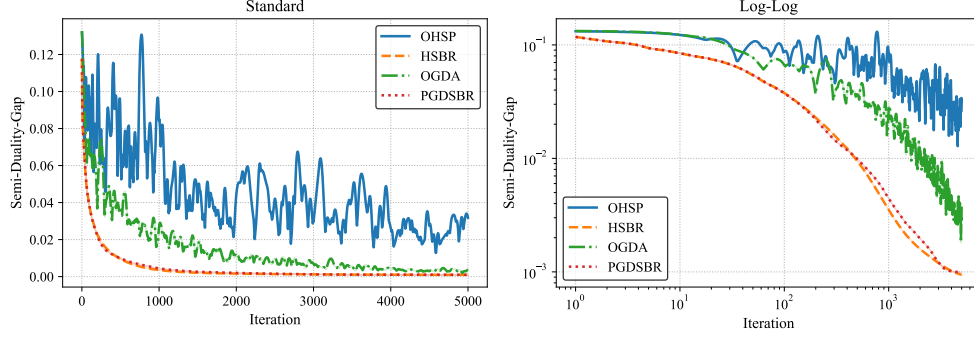


Fig. 3 The comparison of convergence rates among different paradigms.

6 Declarations

6.1 Ethical Statements: it is not applicable.

6.2 Consent to participate: it is not applicable.

6.3 Consent to publish: it is not applicable

6.4 Competing interests

We declare that the authors have no competing interests as defined by Springer, or other interests that might be perceived to influence the results and/or discussion reported in this paper.

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A Mirror Descent in Convex Optimization

This appendix presents key results on mirror descent and its application to convex optimization problems (i.e. Lemma 1). For more details, we refer readers to see the lecture note [37]. As the note considers a more wide scenario, we provide the proofs of the convergence results for completeness.

As ψ is a ρ -strongly convex function in MD, we have

$$D_\psi(x', x) = \psi(x') - \psi(x) - \langle \nabla \psi(x), x' - x \rangle \geq \frac{\rho}{2} \|x' - x\|^2, \quad \forall x, x' \in X, \quad (18)$$

which shows that Bregman divergence can be treated as a pseudo-distance in X . The following *3-Point Property* in [51]¹ will be instrumental in analyzing the properties of the sequence $\{x_t\}$ generated by (MD), which serves as a triangle inequality analog for Bregman divergence:

Lemma 15 (3-Point property) *For any proper lsc convex function $\phi(\cdot)$ and any $z \in X$, let $z^+ := \arg \min_x \{\phi(x) + D_\psi(x, z)\}$, then we have*

$$\phi(x) + D_\psi(x, z) \geq \phi(z^+) + D_\psi(z^+, z) + D_\psi(x, z^+), \quad \forall x \in X. \quad (19)$$

Proof of Lemma 15 Based on the definition of Bregman divergence:

$$\begin{aligned} D_\psi(x, z) &= \psi(x) - \psi(z) - \langle \nabla \psi(z), x - z \rangle, \\ D_\psi(x, z^+) &= \psi(x) - \psi(z^+) - \langle \nabla \psi(z^+), x - z^+ \rangle, \\ D_\psi(z^+, z) &= \psi(z^+) - \psi(z) - \langle \nabla \psi(z), z^+ - z \rangle, \end{aligned}$$

we have

$$D_\psi(x, z) - D_\psi(z^+, z) - D_\psi(x, z^+) = \langle \nabla \psi(z) - \nabla \psi(z^+), z^+ - x \rangle. \quad (20)$$

As z^+ is the minimizer of $\phi(x) + D_\psi(x, z)$, for any $x \in X$, the following holds:

$$\langle y + \nabla \psi(z^+) - \nabla \psi(z), x - z^+ \rangle \geq 0, \quad \forall y \in \partial \phi(z^+).$$

With the convexity of ϕ , we have:

$$\begin{aligned} \langle \nabla \psi(z^+) - \nabla \psi(z), x - z^+ \rangle &\geq \langle y, z^+ - x \rangle & (\forall y \in \partial \phi(z^+)) \\ &\geq \phi(z^+) - \phi(x), \quad \forall x \in X, \end{aligned}$$

which completes the proof with (20). $\square$

Thus Lemma 1 can be obtained:

¹Paul Tseng. Approximation accuracy, gradient methods, and error bound for structured convex optimization. Mathematical Programming, 125(2):263–295, October 2010. ISSN 1436-4646. doi: 10.1007/s10107-010-0394-2.

Proof of Lemma 1 In (MD), $\phi(x) = \frac{1}{\lambda} \langle f'(x_t), x - x_t \rangle$. If f is a L_0 -Lipschitz continuous convex function, then

$$\begin{aligned}
f(x_t) - f(x) &\leq (f'(x_t))^\top (x_t - x) && (f \text{ convex}) \\
&\leq (f'(x_t))^\top (x_t - x_{t+1}) + \lambda D_\psi(x, x_t) - \lambda D_\psi(x, x_{t+1}) - \lambda D_\psi(x_{t+1}, x_t) && (\text{Lemma 15}) \\
&\leq L_0 \|x_t - x_{t+1}\|_2 - \frac{\lambda \rho}{2} \|x_t - x_{t+1}\|_2^2 + \lambda (D_\psi(x, x_t) - D_\psi(x, x_{t+1})), && (f \text{ } L_0\text{-Lipschitz continuous}) \\
&\leq \frac{L_0^2}{2\lambda\rho} + \lambda (D_\psi(x, x_t) - D_\psi(x, x_{t+1})) && (-bz^2 + az \leq \frac{a^2}{4b}, \forall z \in \mathbb{R})
\end{aligned}$$

Summing over t , we obtain:

$$\min_{t \in T} f(x_t) - f(x) \leq \frac{1}{T+1} \sum_{t=0}^T f(x_t) - f(x) \leq \frac{L_0^2}{2\lambda\rho} + \frac{\lambda D_\psi(x, x_0)}{T+1}.$$

If f is L_1 -smooth and $\lambda \geq \frac{L_1}{\rho}$, then

$$\begin{aligned}
f(x_{t+1}) &\leq f(x_t) + \nabla f(x_t)^\top (x_{t+1} - x_t) + \frac{L_1}{2} \|x_{t+1} - x_t\|^2 && (f \text{ } L_1\text{-smooth}) \\
&\leq f(x_t) + \nabla f(x_t)^\top (x_{t+1} - x_t) + \frac{L_1}{\rho} D_\psi(x_{t+1}, x_t) && (\text{Property of Bregman divergence}) \\
&\leq f(x_t) + \nabla f(x_t)^\top (x - x_t) + \lambda (D_\psi(x, x_t) - D_\psi(x, x_{t+1})) + (\frac{L_1}{\rho} - \lambda) D_\psi(x_{t+1}, x_t) && (\text{Lemma 15}) \\
&\leq f(x) + \lambda (D_\psi(x, x_t) - D_\psi(x, x_{t+1})). && (f \text{ convex})
\end{aligned}$$

Let $x = x_t$, then

$$f(x_{t+1}) - f(x_t) \leq -\lambda D_\psi(x_t, x_{t+1}) \leq 0.$$

This demonstrates the monotonic decrease of $f(x_t)$, and the conclusion follows by summation over t .

When f is μ -strongly convex at the same time, the following inequality holds: $\frac{\mu}{2} \|x - x^*\|^2 \leq f(x) - f(x^*)$. Consequently:

$$\frac{\mu}{2} \|x_t - x^*\|^2 \leq f(x_t) - f(x^*) \leq \frac{\lambda D_\psi(x^*, x_0)}{t}.$$

For the Euclidean case $\psi(x) = \frac{1}{2} \|x\|^2$, since $\lambda \geq (\mu + L_1)/2$, applying Lemma 15 with some properties of convex functions in [52] to (MD) yields:

$$\begin{aligned}
D_\psi(x^*, x_{t+1}) &\leq D_\psi(x^*, x_t) + \frac{1}{\lambda} (\nabla f(x_t) - \nabla f(x^*))^\top (x^* - x_{t+1}) - D_\psi(x_{t+1}, x_t) \\
&\leq D_\psi(x^*, x_t) + \frac{1}{\lambda} (\nabla f(x_t) - \nabla f(x^*))^\top (x^* - x_t) + \frac{1}{2\lambda^2} \|\nabla f(x_t) - \nabla f(x^*)\|^2 \\
&\quad + \frac{1}{2} \|x_{t+1} - x_t\|^2 - D_\psi(x_{t+1}, x_t) && (\text{Young's inequality}) \\
&\leq D_\psi(x^*, x_t) + \frac{1}{\lambda} (\nabla f(x_t) - \nabla f(x^*))^\top (x^* - x_t) + \frac{1}{2\lambda^2} \|\nabla f(x_t) - \nabla f(x^*)\|^2
\end{aligned}$$

$$\begin{aligned}
&\leq D_\psi(x^*, x_t) - \frac{1}{\lambda} \frac{\mu L_1}{\mu + L_1} \|x_t - x^*\|^2 + \left(\frac{1}{2\lambda^2} - \frac{1}{\lambda} \frac{1}{\mu + L_1}\right) \|\nabla f(x_t) - \nabla f(x^*)\|^2 \\
&\quad \text{(Theorem 2.1.12 in [52])} \\
&\leq D_\psi(x^*, x_t) + \left(\mu^2 \left(\frac{1}{2\lambda^2} - \frac{1}{\lambda} \frac{1}{\mu + L_1}\right) - \frac{1}{\lambda} \frac{\mu L_1}{\mu + L_1}\right) \|x_t - x^*\|^2. \\
&\quad \text{(Theorem 2.1.10 in [52])}
\end{aligned}$$

Since $D_\psi(x, y) = \frac{1}{2} \|x - y\|^2$, we obtain:

$$\|x_{t+1} - x^*\|^2 \leq \left(1 - \frac{\mu}{\lambda}\right)^2 \|x_t - x^*\|^2,$$

leading to the exponential convergence rate:

$$\|x_t - x^*\| \leq \left(1 - \frac{\mu}{\lambda}\right)^t \|x_0 - x^*\|.$$

□

B Missing Examples, Lemmas and Proofs in Section 3

B.1 Additional materials for Section 3.1

B.1.1 Proofs of Theorem 3

Proof of Lemma 2 Using Lemma 15 with substitutions: $x_t = z$, $x_{t+1} = z^+$, $\phi(w) = \frac{1}{\lambda} \langle Ay_t, w - x_t \rangle$, we obtain the following inequality:

$$\begin{aligned}
D_\psi(x, x_{t+1}) &\leq D_\psi(x, x_t) + \frac{1}{\lambda} \langle Ay_t, x - x_t + x_t - x_{t+1} \rangle - D_\psi(x_{t+1}, x_t) \\
&\leq D_\psi(x, x_t) + \frac{1}{\lambda} \langle Ay_t, x - x_t \rangle + \frac{\|Ay_t\| \|x_t - x_{t+1}\|}{\lambda} - D_\psi(x_{t+1}, x_t).
\end{aligned}$$

Specializing to $x = x_t$ and leveraging the ρ -strong convexity of ψ yield $D_\psi(x_t, x_{t+1}) + D_\psi(x_{t+1}, x_t) \leq \|Ay_t\| \|x_t - x_{t+1}\| / \lambda$. By the definition of D_ψ , we have $D_\psi(x_{t+1}, x_t) \geq \frac{\rho}{2} \|x_t - x_{t+1}\|^2$. Combining this two results implies $\|x_t - x_{t+1}\| \leq \frac{\|Ay_t\|}{\lambda}$. Then let $x = x^* \in X^*$, we have:

$$D_\psi(x^*, x_{t+1}) \leq D_\psi(x^*, x_t) + \frac{1}{\lambda} \langle Ay_t, x^* - x_t \rangle + \frac{\|Ay_t\|^2}{2\rho\lambda}.$$

□

Lemma 16 *The function V is convex and $\sqrt{n}\|A\|_{\max}$ -Lipschitz continuous.*

Proof of Lemma 16 First, for any $x, x' \in \Delta(\mathcal{I})$ and $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha)x' \in \Delta(\mathcal{I})$. Let $y^\alpha \in \text{BR}(\alpha x + (1 - \alpha)x')$. Then, we have

$$\begin{aligned}
V(\alpha x + (1 - \alpha)x') &= (\alpha x + (1 - \alpha)x')^\top Ay^\alpha \\
&= \alpha x^\top Ay^\alpha + (1 - \alpha)(x')^\top Ay^\alpha \\
&\leq \alpha \max_{y \in \Delta(\mathcal{J})} x^\top Ay + (1 - \alpha) \max_{y \in \Delta(\mathcal{J})} (x')^\top Ay \\
&= \alpha V(x) + (1 - \alpha)V(x').
\end{aligned}$$

Therefore, V is convex.

Then, we prove the Lipschitz continuity of V . For any $y \in \Delta(\mathcal{J})$, we have $\|Ay\| = (\sum_{i=1}^n (\sum_{j=1}^m a_{ij}y_j)^2)^{1/2} \leq \sqrt{n}\|A\|_{\max}$. For any $x, x' \in \Delta(\mathcal{I})$, let $y^x \in \text{BR}(x)$ and $y^{x'} \in \text{BR}(x')$. On one hand,

$$\begin{aligned} V(x) - V(x') &= x^\top Ay^x - (x')^\top Ay^{x'} \\ &= (x^\top Ay^x - x^\top Ay^{x'}) + (x^\top Ay^{x'} - (x')^\top Ay^{x'}) \\ &\geq x^\top Ay^{x'} - (x')^\top Ay^{x'} \\ &\geq -\|x - x'\| \|Ay^{x'}\| \\ &\geq -\sqrt{n}\|A\|_{\max}\|x - x'\|. \end{aligned}$$

On the other hand,

$$\begin{aligned} V(x) - V(x') &= x^\top Ay^x - (x')^\top Ay^{x'} \\ &= (x^\top Ay^x - (x')^\top Ay^x) + ((x')^\top Ay^x - (x')^\top Ay^{x'}) \\ &\leq x^\top Ay^x - (x')^\top Ay^x \\ &\leq \|x - x'\| \|Ay^x\| \\ &\leq \sqrt{n}\|A\|_{\max}\|x - x'\|. \end{aligned}$$

In conclusion, $|V(x) - V(x')| \leq \sqrt{n}\|A\|_{\max}\|x - x'\|$. Hence, the function V is $\sqrt{n}\|A\|_{\max}$ -Lipschitz continuous. $\square$

Lemma 17 For any $y^x \in \text{BR}(x)$, Ay^x is a subgradient of function V at x .

Proof of Lemma 17 Given that $y^x \in \text{BR}(x)$, then for any $x' \in \Delta(\mathcal{I})$, we have that

$$V(x') = \max_{y \in \Delta(\mathcal{J})} (x')^\top Ay \geq (x')^\top Ay^x = x^\top Ay^x + \langle Ay^x, x' - x \rangle = V(x) + \langle Ay^x, x' - x \rangle,$$

which means that Ay^x is a subgradient of V at x . $\square$

Another Proof of Theorem 3 Using Lemma 1 Since Player Y takes a BR to the stage strategy of Player X at each stage, i.e., $y_t \in \text{BR}(x_t)$, by Lemma 17, Ay_t is a subgradient of V at x_t . Thus, the update rule of Player X in Equation (6) can be rewritten as

$$x_{t+1} = \arg \min_{x \in \Delta(\mathcal{I})} \{ \langle V'(x_t), x - x_t \rangle + \lambda_t D_\psi(x, x_t) \}, \quad t \geq 0.$$

This implies that when Player Y adopts BR, the update of the stage strategy of Player X essentially corresponds to using MD to iteratively minimize the function V , whose minimum represents the value of the game. Based on the convergence result of MD in Lemma 1, we prove Theorem 3. $\square$

Lemma 18 (Hoffman [43]) Given a finite consistent system of linear inequalities

$$Ax \leq b, \tag{21}$$

where A is an $m \times n$ real matrix, $x \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$. Then there exists a constant $c > 0$, such that for any $y \in \mathbb{R}^n$ there exists a solution x^* of (21) satisfying:

$$\|y - x^*\| \leq c\|(Ax - b)^+\|_\infty,$$

where $a^+ = (a + |a|)/2$.

B.1.2 Omitted proofs of Theorem 12

For completeness, we include a statement of the envelope theorem below. By the envelope theorem [47], we directly show that $Ay(x; \tau)$ corresponds precisely to the gradient of the function V . Then, we can also prove the convexity of $V(x; \tau)$.

Lemma 19 (Envelope Theorem) *Suppose that X is a convex set and $f : X \times [0, 1] \rightarrow \mathbb{R}^n$ is a concave function. Also suppose that $t_0 \in (0, 1)$, and that there is some $x^* \in X^*(t_0) = \{x \in X : f(x, t) = \sup_{x \in X} f(x, t)\}$ such that $f_t(x^*, t_0)$ exists. Then $V(t) = \sup_{x \in X} f(x, t)$ is differentiable at t_0 and $V'(t_0) = f_t(x^*, t_0)$.*

Proof of Lemma 8 The first part follows immediately from Lemma 19. For the convexity of $V(x; \tau)$, for any $x, x' \in \Delta(\mathcal{I})$,

$$\begin{aligned} & \langle \nabla_x V(x; \tau) - \nabla_x V(x'; \tau), x - x' \rangle \\ &= \langle Ay(x; \tau) - Ay(x'; \tau), x - x' \rangle \\ &= x^\top Ay(x; \tau) - \tau h(y(x; \tau)) - x'^\top Ay(x'; \tau) + \tau h(y(x'; \tau)) \\ &\quad - (x')^\top Ay(x; \tau) + \tau h(y(x; \tau)) + (x')^\top Ay(x'; \tau) - \tau h(y(x'; \tau)) \\ &\geq 0, \end{aligned}$$

where the last inequality follows by the definition of $y(x; \tau)$ and $y(x'; \tau)$. Since $\Delta(\mathcal{I})$ is convex, the function $V(x; \tau)$ is convex on $\Delta(\mathcal{I})$. $\square$

Proof of Lemma 9 Since $y(x; \tau)$ is the unique solution to the constrained optimization problem $\max_{y \in \Delta(\mathcal{J})} \{x^\top Ay - \tau h(y)\}$, the optimality conditions yield that

$$\langle A^\top x - \tau \nabla h(y(x; \tau)), y(x'; \tau) - y(x; \tau) \rangle \leq 0,$$

and

$$\langle A^\top x' - \tau \nabla h(y(x'; \tau)), y(x; \tau) - y(x'; \tau) \rangle \leq 0.$$

The K -strong convexity of the function h gives that

$$\langle \nabla h(y(x; \tau)) - \nabla h(y(x'; \tau)), y(x; \tau) - y(x'; \tau) \rangle \geq K \|y(x; \tau) - y(x'; \tau)\|^2.$$

Combining these three inequalities, we obtain that

$$\begin{aligned} \tau K \|y(x; \tau) - y(x'; \tau)\|^2 &\leq \langle A^\top (x - x'), y(x; \tau) - y(x'; \tau) \rangle \\ &\leq \|A\| \|x - x'\| \|y(x; \tau) - y(x'; \tau)\|. \end{aligned}$$

Consequently, $\|y(x; \tau) - y(x'; \tau)\| \leq \frac{\|A\|}{\tau K} \|x - x'\|$, implying the Lipschitz continuity of $y(x; \tau)$ in x .

Furthermore, when $h(y) = \sum_{j=1}^m h_j(y_j)$ and $y(x; \tau)$ lies in the interior of $\Delta(\mathcal{J})$ for all $x \in \Delta(\mathcal{I})$, consider the Lagrangian function:

$$L(x, y, \alpha, \beta) = -x^\top Ay + \tau \sum_{j=1}^m h_j(y_j) + \alpha \left(\sum_{j=1}^m y_j - 1 \right) - \sum_{j=1}^m \beta_j y_j$$

The corresponding KKT conditions are

$$\begin{cases} F_j(x, y, \alpha, \beta) = -\sum_{i=1}^n a_{ij}x_i + \tau \frac{\partial h_j}{\partial y_j}(y) + \alpha - \beta_j = 0, & \forall j = 1, \dots, m, \\ F_{m+j}(x, y, \alpha, \beta) = \beta_j y_j = 0, & \forall j = 1, \dots, m, \\ F_{2m+1}(x, y, \alpha, \beta) = \sum_{j=1}^m y_j - 1 = 0, \\ \alpha \geq 0, \beta_j \geq 0, y_j \geq 0, & \forall j = 1, \dots, m. \end{cases} \quad (22)$$

The Jacobian determinant is:

$$\begin{aligned} \det \left(\frac{\partial F}{\partial(y, \alpha, \beta)} \right) &= \det \begin{pmatrix} \tau \operatorname{diag}(\tilde{h}) & -I & e \\ \operatorname{diag}(\beta) & \operatorname{diag}(y) & 0 \\ e^\top & 0 & 0 \end{pmatrix} \\ &= -\tau^{m-1} \left(\sum_{j=1}^m \frac{1}{\tilde{h}_j} \right) \left(\prod_{j=1}^m \tilde{h}_j \right) \left(\prod_{j=1}^m \left(\left(y_j + \frac{\beta_j}{\tau \tilde{h}_j} \right) + \frac{\beta_j}{\tau \tilde{h}_j} \right) \right). \end{aligned}$$

where $\tilde{h}_j = h_j''(y_j) > 0$ and $e = (1, \dots, 1)^\top$. Since $y(x; \tau)$ lies in the interior of $\Delta(\mathcal{J})$, we obtain that $\beta_j = 0$ for all j and the determinant is nonzero. By the implicit function theorem, differentiating (22) yields:

$$\begin{cases} -\sum_{i=1}^n a_{ij}dx_i + \tau \tilde{h}_j dy_j + d\alpha - d\beta_j = 0, & \forall j = 1, \dots, m, \\ \sum_{j=1}^m dy_j = 0, \\ d\beta_j = 0, & \forall j = 1, \dots, m, \end{cases}$$

This leads to the block matrix equation:

$$\begin{pmatrix} \tau \operatorname{diag}(\tilde{h}) & e \\ e^\top & 0 \end{pmatrix} \begin{pmatrix} dy \\ d\alpha \end{pmatrix} = \begin{pmatrix} A^\top & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} dx \\ 0 \end{pmatrix}.$$

The solution of this equation is $dy = \frac{1}{\tau} X A^\top dx$, where X is a symmetric matrix with elements $X_{ij} = \delta_{ij}/\tilde{h}_i - (\tilde{h}_i \tilde{h}_j \sum_{k=1}^m \tilde{h}_k^{-1})^{-1}$. By the Gershgorin circle theorem, $\|X\|$ is bounded by $\max_{1 \leq j \leq m} \sup_{x \in \Delta(\mathcal{J})} (h_j''(y_j(x; \tau)))^{-1} = M_{\tilde{h}}$, and thus $y(x; \tau)$ is $\frac{\|A\| M_{\tilde{h}}}{\tau}$ -Lipschitz continuous. $\square$

Proof of Lemma 11 Let $x_\tau^* \in X^*(\tau)$ with $y^\tau \in \operatorname{BR}(x_\tau^*)$ and x^* be a NE strategy of Player X with $y^* \in \operatorname{BR}(x^*)$, thus $V(x^*) = v^*$. On the one hand, it is obvious that $V(x_\tau^*) - V(x^*) \geq 0$ since x^* is a minima of V . On the other hand,

$$\begin{aligned} &V(x_\tau^*) - V(x^*) \\ &= (x_\tau^*)^\top A y^\tau - (x^*)^\top A y^* \\ &\leq (x_\tau^*)^\top A y^\tau - (x^*)^\top A y(x^*; \tau) \\ &= ((x_\tau^*)^\top A y^\tau - \tau h(y^\tau)) - ((x^*)^\top A y(x^*; \tau) - \tau h(y(x^*; \tau))) + \tau h(y^\tau) - \tau h(y(x^*; \tau)) \\ &\leq V(x_\tau^*; \tau) - V(x^*; \tau) + 2\tau M_h \\ &\leq 2\tau M_h \end{aligned}$$

This completes the proof of the first inequality. And given any $x \in \Delta(\mathcal{I})$, let $y^x \in \operatorname{BR}(x)$. Then,

$$\begin{aligned} V(x; \tau) - V(x) &= x^\top A y(x; \tau) - \tau h(y(x; \tau)) - x^\top A y^x \\ &\leq -\tau h(y(x; \tau)) \\ &\leq \tau M_h \end{aligned}$$

Conversely,

$$\begin{aligned}
V(x) - V(x; \tau) &= x^\top Ay^x - (x^\top Ay(x; \tau) - \tau h(y(x; \tau))) \\
&= (x^\top Ay^x - \tau h(y^x)) - (x^\top Ay(x; \tau) - \tau h(y(x; \tau))) + \tau h(y^x) \\
&\leq \tau h(y^x) \\
&\leq \tau M_h.
\end{aligned}$$

Combining these two inequalities yields $|V(x; \tau) - V(x)| \leq \tau M_h$. $\square$

B.2 Additional materials for Section 3.3

Lemma 20 *Function $V(x; \epsilon, \tau)$ is ϵ -strongly convex and L_2 -smooth, where $L_2 = \frac{\|A\|^2}{\tau K} + \epsilon$.*

Proof of Lemma 20 By the envelope theorem, we have $\nabla_x V(x; \epsilon, \tau) = Ay(x; \tau) + \epsilon x$. Then, for any $x, x' \in \Delta(\mathcal{I})$, we have

$$\begin{aligned}
&\langle \nabla_x V(x; \epsilon, \tau) - \nabla_x V(x'; \epsilon, \tau), x - x' \rangle \\
&= \langle Ay(x; \tau) + \epsilon x - Ay(x'; \tau) - \epsilon x', x - x' \rangle \\
&= \langle Ay(x; \tau) - Ay(x'; \tau), x - x' \rangle + \epsilon \|x - x'\|^2 \\
&= x^\top Ay(x; \tau) - x^\top Ay(x'; \tau) + (x')^\top Ay(x'; \tau) - (x')^\top Ay(x; \tau) + \epsilon \|x - x'\|^2 \\
&= x^\top Ay(x; \tau) - \tau h(y(x; \tau)) - x^\top Ay(x'; \tau) + \tau h(y(x'; \tau)) \\
&\quad + (x')^\top Ay(x'; \tau) - \tau h(y(x'; \tau)) - (x')^\top Ay(x; \tau) + \tau h(y(x; \tau)) + \epsilon \|x - x'\|^2 \\
&\geq \epsilon \|x - x'\|^2.
\end{aligned}$$

By the definition of strong convexity, the function $V(x; \epsilon, \tau)$ is ϵ -strongly convex.

By Lemma 9, function $y(x; \tau)$ is $\frac{\|A\|}{\tau K}$ -Lipschitz continuous with respect to x , thus

$$\begin{aligned}
&\|\nabla_x V(x; \epsilon, \tau) - \nabla_x V(x'; \epsilon, \tau)\| \\
&= \|Ay(x; \tau) + \epsilon x - Ay(x'; \tau) - \epsilon x'\| \\
&\leq \|Ay(x; \tau) - Ay(x'; \tau)\| + \|\epsilon x - \epsilon x'\| \\
&\leq \|A\| \|y(x; \tau) - y(x'; \tau)\| + \epsilon \|x - x'\| \\
&\leq \frac{\|A\|^2}{\tau K} \|x - x'\| + \epsilon \|x - x'\| \\
&= \left(\frac{\|A\|^2}{\tau K} + \epsilon \right) \|x - x'\|.
\end{aligned}$$

This implies that the function $V(x; \epsilon, \tau)$ is $(\frac{\|A\|^2}{\tau K} + \epsilon)$ -smooth. $\square$

Lemma 21 *For any $x \in \Delta(\mathcal{I})$ and $\tau, \epsilon > 0$, we have*

$$\begin{aligned}
-\tau M_h &\leq V(x; \epsilon, \tau) - V(x) \leq \frac{\epsilon}{2} + \tau M_h. \\
0 &\leq V(x^*(\epsilon, \tau)) - v^* \leq \frac{\epsilon}{2} + 2\tau M_h.
\end{aligned}$$

Proof of Lemma 21 On one hand, for any $y^x \in \text{BR}(x)$,

$$\begin{aligned}
& V(x; \epsilon, \tau) - V(x) \\
&= x^\top Ay(x; \tau) + \frac{\epsilon}{2} \|x\|^2 - \tau h(y(x; \tau)) - x^\top Ay^x \\
&= x^\top Ay(x; \tau) - \tau h(y(x; \tau)) - x^\top Ay^x + \tau h(y^x) - \tau h(y^x) + \frac{\epsilon}{2} \|x\|^2 \\
&\geq -\tau h(y^x) + \frac{\epsilon}{2} \|x\|^2 \\
&\geq -\tau M_h.
\end{aligned}$$

On the other hand, for any $y^x \in \text{BR}(x)$,

$$\begin{aligned}
& V(x; \epsilon, \tau) - V(x) \\
&= x^\top Ay(x; \tau) + \frac{\epsilon}{2} \|x\|^2 - \tau h(y(x; \tau)) - x^\top Ay^x \\
&\leq \frac{\epsilon}{2} \|x\|^2 - \tau h(y(x; \tau)) \\
&\leq \frac{\epsilon}{2} + \tau M_h.
\end{aligned}$$

This completes the first inequality. And for the second inequality, it is obvious that $V(x^*(\epsilon, \tau)) - v^* \geq 0$. And for any $y^\epsilon \in \text{BR}(x^*(\epsilon, \tau))$ and $y^* \in \text{BR}(x^*)$,

$$\begin{aligned}
& V(x^*(\epsilon, \tau)) - v^* \\
&= (x^*(\epsilon, \tau))^\top Ay^\epsilon - (x^*)^\top Ay^* \\
&= (x^*(\epsilon, \tau))^\top Ay^\epsilon - \tau h(y^\epsilon) - (x^*)^\top Ay^* + \tau h(y^\epsilon) \\
&\leq (x^*(\epsilon, \tau))^\top Ay(x^*(\epsilon, \tau); \tau) - \tau h(y(x^*(\epsilon, \tau); \tau)) - (x^*)^\top Ay^* + \tau M_h \\
&= (x^*(\epsilon, \tau))^\top Ay(x^*(\epsilon, \tau); \tau) + \frac{\epsilon}{2} \|x^*(\epsilon, \tau)\|^2 - \tau h(y(x^*(\epsilon, \tau); \tau)) \\
&\quad - (x^*)^\top Ay^* + \tau M_h - \frac{\epsilon}{2} \|x^*(\epsilon, \tau)\|^2 \\
&\leq (x^*)^\top Ay(x^*; \tau) + \frac{\epsilon}{2} \|x^*\|^2 - \tau h(y(x^*; \tau)) - (x^*)^\top Ay^* - \frac{\epsilon}{2} \|x^*(\epsilon, \tau)\|^2 + \tau M_h \\
&\leq \frac{\epsilon}{2} \|x^*\|^2 - \tau h(y(x^*; \tau)) - \frac{\epsilon}{2} \|x^*(\epsilon, \tau)\|^2 + \tau M_h \\
&\leq \frac{\epsilon}{2} + 2\tau M_h.
\end{aligned}$$

□